

Relative Wealth Concerns, Network Aggregation, and Effective Risk Aversion

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Abstract

In a CARA economy where each investor compares his wealth to a weighted average over a network of peers, equilibrium demand scales solve a Katz–Bonacich system. We characterise how the network aggregates risk aversion through two limits that hold for every network. At zero comparison intensity the marginal effect on the effective coefficient is a quadratic form in risk tolerances; when attention is balanced, in the sense that the influence matrix is doubly stochastic, that form is a Dirichlet form and the effect is non-negative, vanishing only if tolerance is constant across links. Under intense comparison the effective coefficient converges to a stationary average of individual coefficients, weighted by the stationary distribution of the influence matrix rather than by population. The two limits organise the rest. Comparison lowers the effective coefficient whenever stationary influence favours the less risk averse, which explains sign reversal; it interpolates from the harmonic to the arithmetic mean when influence is spread evenly, which is the mean-field case; and it does nothing at all when the network splits into blocks that are homogeneous in type, which is perfect homophily. A separate result is that the comparison is exactly neutral, for every network and every profile of intensities, when net risk aversion is common.

Keywords: relative wealth concerns, keeping up with the Joneses, CARA aggregation, Katz–Bonacich centrality, homophily.

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1 Introduction

[12] studies an exchange economy in which agents evaluate consumption relative to a weighted average of the consumption of others, and asks when equilibrium prices depend

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only on aggregates. His Proposition 1 shows that linear risk tolerance with common cautiousness delivers a representative agent and rules out portfolio bias; his Proposition 2 gives the conditions under which prices depend on the aggregate endowment alone, and finds them restrictive: the preference parameters must be common across agents and, absent the trivial case of uniform weights, so must the equilibrium value of the endowment. Under those conditions the CARA case reduces cleanly, and keeping up with the Joneses is equivalent to lowering the representative agent's risk aversion from α to $\alpha(1 - \gamma)$.

What happens outside those conditions is the natural question, and [12] leaves it open. He notes that in their absence equilibrium prices may be affected by the distribution of weights in the definition of the Joneses, by the preference parameters, and by the endowment distribution — three channels that a representative-agent reduction cannot express. This paper opens the first of them, in the CARA case, by letting the benchmark be agent-specific: investor a compares himself to $\sum_j A_{aj}W_j$, a weighted average over a network of influence rather than a single economy-wide aggregate. The question is then not whether the comparison matters but *what feature of the network* decides how it matters.

Two remarks on what is being held fixed, since they govern the reading of everything below. First, with preferences $-\exp[-\theta_a W_a - \lambda_a(W_a - \sum_j A_{aj}W_j)]$, the coefficient of absolute risk aversion in own wealth is $\alpha_a = \theta_a + \lambda_a$, not θ_a . We call θ_a *net* risk aversion: it is the coefficient on a common translation of all wealths, and it is risk aversion net of the comparison. Raising λ_a at fixed θ_a therefore raises own-wealth risk aversion too, and the comparative statics below are taken along that path. Second, market clearing here delivers a mean-covariance relation conditional on the covariance matrix, not a complete system of asset prices; the object we characterise is the effective market coefficient $\bar{\theta}$, and we claim nothing about price levels.

The demand scales solve $[\mathbb{D}_{\theta+\lambda} - \mathbb{D}_\lambda A]\vec{c} = \mathbf{1}$, a Katz-Bonacich system in the sense of [2], and the paper's contribution is to characterise how that system aggregates. Two limits, both available for every network, do the work.

At the origin, the marginal effect of comparison is the quadratic form $\bar{\theta}'(0) = m \vec{t}^\top (\mathbb{I} - A) \vec{t} / (\mathbf{1}^\top \vec{t})^2$ in the vector $\vec{t}$ of risk tolerances (Theorem 2). If attention is balanced — A doubly stochastic, so that each investor receives as much attention as he gives, and not necessarily symmetric — this is a Dirichlet form, $\frac{1}{2} \sum_{a,j} A_{aj} (t_a - t_j)^2$, hence non-negative and zero only if tolerance is constant along every link. Comparison raises the effective coefficient exactly to the extent that investors look at people unlike themselves.

At the other end, as comparison becomes intense, $\bar{\theta}$ converges to a *stationary* average of individual coefficients: $\vec{\pi}^\top \vec{\theta}$ when the influence matrix has a single recurrent class, and a harmonic combination across classes in general (Theorem 3). Influence, not population, does the weighting in the limit, and since the no-comparison value is the harmonic mean $H(\theta)$, a sufficient condition for comparison to *lower* the effective coefficient is $\vec{\pi}^\top \vec{\theta} < H(\theta)$.

These two limits organise what would otherwise be a list. The mean-field regime is the case in which the stationary distribution is uniform, so the limit is $\mathbb{E}[\theta]$ and

comparison interpolates from the harmonic to the arithmetic mean (Proposition 1). Sign reversal is the case in which stationary influence favours the less risk averse; in the three-investor example of Proposition 2 the most risk averse investor is watched by nobody, receives stationary weight zero, and the limit falls to $3/4$ against a starting value of $6/7$. Perfect homophily is the case of several recurrent classes each homogeneous in type, where the stationary average reproduces the harmonic mean and the two limits coincide, so the aggregate never moves at all (Proposition 3).

A separate and much simpler result delimits where any of this can happen. If net risk aversion is common, the comparison is exactly neutral: every investor holds the position he would hold with no relative concern, for every row-stochastic A and every profile λ_a (Theorem 1). No statistic of the network survives into equilibrium. This is the network extension of Gómez’s CARA case, and it says that dispersion of net risk aversion is the necessary fuel for everything else.

The message is that the comparison channel in this class of models is a property neither of preferences alone nor of the network alone, but of the alignment between them — specifically, of the correlation between net risk aversion and stationary influence. A representative-agent treatment delivers the mean-field case, and can therefore be wrong in sign, not merely in magnitude, whenever attention is unbalanced.

2 Related literature

Preferences over relative consumption or wealth have been used to address a range of asset-pricing questions. [1] and [10] build consumption-based models with external benchmarks and obtain an amplified equity premium; [3] show how an external habit generates time-varying effective risk aversion; [4] make the wealth distribution a state variable under heterogeneous CRRA preferences, which is the dynamic counterpart of the configuration studied here. [9] provide the taxonomy under which (1) is jealousy combined with keeping up with the Joneses.

Within the CARA class, [5] is the closest antecedent and the comparison locates what is new here. They study two agents with multiple risky assets and distributions in a natural exponential family, and show that relative concerns act through a transformation of effective risk tolerances, lowering equilibrium premia. The mechanism they identify is a change in the tolerance *levels*. With two agents the benchmark of each is the other, so there is no network to speak of: the influence matrix is forced to $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is doubly stochastic, and by Theorem 2 the marginal effect is then necessarily non-negative. The questions of this paper — whether attention is balanced, which investors stationary influence favours, whether types and positions are correlated — cannot be posed at $m = 2$, and the results that turn on them (Theorems 2 and 3, and the reversal and homophily that follow) have no two-agent counterpart. What is added is therefore not a dimension count but the object doing the aggregating: a stationary distribution in place of a pair of tolerances. [8] obtain portfolio bias when the benchmark differs in composition from the investor’s own holdings, a channel excluded here by the scalar structure of (4). [14] solve relative-performance games for CARA and CRRA investors

in mean-field and n -agent form; their concern is the existence and structure of Nash equilibria rather than whether the aggregate moves, and their mean-field formulation is the analogue of our Proposition 1 in a different setting. [13] take the problem to incomplete markets in continuous time. [16] surveys what heterogeneity and inequality imply for asset prices more broadly.

Two connections deserve care. [11] gives the criterion governing the effect of endowment inequality on the equity premium, namely concavity of risk tolerance in wealth. Our (14) is an analogue for dispersion in a preference parameter rather than in endowments, and should be read as such rather than as an application of his theorem. And [2] give the linear-quadratic network game whose solution (5) reproduces; what the present setting adds is that the centrality vector becomes constant, and every statistic of the network drops out, precisely when net risk aversion is common.

3 The economy

There are m investors, one riskless asset with net return r_f , and n risky assets with excess returns $\vec{y}$ that are Gaussian with mean $\vec{r}_e$ and non-singular covariance Σ . Investor a begins with wealth W_a^0 and chooses monetary positions $\vec{\pi}_a \in \mathbb{R}^n$, so that $W_a = W_a^0(1 + r_f) + \vec{\pi}_a^\top \vec{y}$. Preferences are

$$U_a = -\exp \left[-\theta_a W_a - \lambda_a \left(W_a - \sum_j A_{aj} W_j \right) \right], \quad \theta_a > 0, \lambda_a \geq 0. \quad (1)$$

Assumption 1 (Normalised benchmark). $m \geq 2$; $A_{aa} = 0$, $A_{aj} \geq 0$ and $\sum_j A_{aj} = 1$ for every a .

Definition 1 (Two coefficients). Holding the wealth of others fixed, the coefficient of absolute risk aversion of (1) in own wealth is

$$\alpha_a = \theta_a + \lambda_a, \quad (2)$$

which we call *own-wealth risk aversion*. We call θ_a the *net risk-aversion coefficient*: under Assumption 1 it is the coefficient on a common translation of all wealths, since adding a constant to every W_j shifts the exponent of (1) by $-\theta_a$ times that constant, independently of λ_a . It is risk aversion net of the comparison.

The distinction matters for reading everything below, and we state it before any result because the two coefficients cannot both be held fixed. Raising λ_a at constant θ_a is not a pure increase in the intensity of comparison: by (2) it raises own-wealth risk aversion by the same amount. Every comparative static in λ in this paper is taken along that path, at constant θ , and the neutrality of Section 4 is exactly the cancellation of the two effects. The alternative path, at constant α , is described in Remark 2.

The multiplicative parametrization of [12] makes the same point in his notation. Collecting terms in the exponent of (1),

$$-\theta_a W_a - \lambda_a \left(W_a - \sum_j A_{aj} W_j \right) = -\alpha_a \left(W_a - \gamma_a \sum_j A_{aj} W_j \right), \quad \gamma_a = \frac{\lambda_a}{\theta_a + \lambda_a} \in [0, 1), \quad (3)$$

so that $\theta_a = \alpha_a(1 - \gamma_a)$. He writes the CARA case as $u(c, X) = -\exp(-\alpha(c - \gamma X))$ and reduces it to a representative agent with coefficient $\alpha(1 - \gamma)$, which is exactly θ .

The benchmark is thus a weighted average of the wealth of the other investors, as in [12], with the difference that the weights may vary across investors. Assumption 1 is what gives θ_a the reading in Definition 1, and it is what the results below require. Dropping it changes the object being compared: row sums below one describe comparison with a partial or external benchmark, which is a coherent alternative we do not study, while row sums above $1 + \theta_a/\lambda_a$ make an investor worse off when the whole economy becomes richer, which is harder to motivate. We retain Assumption 1 throughout and claim nothing about either case.

Assumption 2 (Nash behaviour). Each investor chooses $\vec{\pi}_a$ taking the positions $\vec{\pi}_j$, $j \neq a$, as fixed at their equilibrium values. An equilibrium is a profile $(\vec{\pi}_1, \dots, \vec{\pi}_m)$ and a vector $\vec{r}_e$ such that each $\vec{\pi}_a$ is optimal given the others and markets clear.

Assumption 2 is the standard notion and we state it explicitly because the conjecture it embodies matters for what follows. An investor behaving in this way conjectures that the positions of others do not respond to his own: the conjectural derivative is zero, and the conjectured levels are the equilibrium positions.

First-order conditions

Under (1) the random part of the exponent is $-(\theta_a + \lambda_a)\vec{\pi}_a - \lambda_a \sum_j A_{aj}\vec{\pi}_j^\top \vec{y}$. Writing $\vec{q}_a$ for the bracket, Gaussian returns give $\mathbb{E}[U_a] = -C \exp(-\vec{q}_a^\top \vec{r}_e + \frac{1}{2}\vec{q}_a^\top \Sigma \vec{q}_a)$, whose exponent is strictly convex in $\vec{\pi}_a$ because $\partial \vec{q}_a / \partial \vec{\pi}_a = (\theta_a + \lambda_a)\mathbb{I}$ is non-singular. The optimum is interior and solves $\vec{q}_a = \Sigma^{-1}\vec{r}_e$, that is

$$(\theta_a + \lambda_a)\vec{\pi}_a - \lambda_a \sum_j A_{aj}\vec{\pi}_j = \Sigma^{-1}\vec{r}_e. \quad (4)$$

Every solution is proportional to $\Sigma^{-1}\vec{r}_e$, so all investors hold the same portfolio of risky assets and differ only in scale; two-fund separation holds, as in Gómez's Proposition 1. Writing $\vec{\pi}_a = c_a \Sigma^{-1}\vec{r}_e$ reduces (4) to

$$[\mathbb{D}_{\theta+\lambda} - \mathbb{D}_\lambda A] \vec{c} = \mathbf{1}, \quad \mathbb{D}_x = \text{diag}(x_1, \dots, x_m), \quad (5)$$

whose solution is a Katz–Bonacich centrality of the influence network [2], generalised in that the attenuation factors λ_a and the base vector are agent-specific rather than scalar. Write $M = \mathbb{D}_{\theta+\lambda} - \mathbb{D}_\lambda A$. Its off-diagonal entries are non-positive, so M is a Z-matrix,

and its off-diagonal row sums are $\lambda_a \sum_j A_{aj} = \lambda_a < \theta_a + \lambda_a$, so it is strictly diagonally dominant with positive diagonal. A strictly diagonally dominant Z-matrix with positive diagonal is a non-singular M-matrix, hence $M^{-1} \geq 0$ entrywise [7]; existence and uniqueness follow, and $\vec{c} = M^{-1}\mathbf{1} > 0$ because no row of M^{-1} can vanish. Market clearing against a supply with value weights $\vec{w} \geq 0$, $\sum_i w_i = 1$, and aggregate wealth $W_M^0 = \sum_a W_a^0$ measured in the same units as W_a^0 , gives

$$\vec{r}_e = \theta_M W_M^0 \Sigma \vec{w}, \quad \frac{1}{\theta_M} = \sum_a c_a, \quad \bar{\theta} = m \theta_M. \quad (6)$$

We call $\bar{\theta}$ the effective market risk-aversion coefficient and reserve *mean-covariance price-of-risk coefficient* for the product $W_M^0 \theta_M = \bar{W}^0 \bar{\theta}$, which is the dimensionless object; $\bar{\theta}$ alone carries units of inverse wealth and we do not call it a price.

Remark 1 (What the model determines). Equations (4)–(6) determine a mean-covariance relation given Σ and the supplies. They are not a complete system of asset prices: with payoffs and quantities as primitives a change in prices would move both the mean and the covariance of returns. Statements below concern demands and expected excess returns conditional on Σ , and nothing is claimed about price levels, market volatility or the invariance of betas.

4 Neutrality under common net risk aversion

Theorem 1. *Let Assumptions 1 and 2 hold and suppose $\theta_a = \theta$ for every a , that is, $\alpha_a(1 - \gamma_a) = \theta$. Then $\vec{c} = \theta^{-1}\mathbf{1}$ is the unique solution of (5), so every investor holds $\vec{\pi}_a = \Sigma^{-1}\vec{r}_e/\theta$ and $\bar{\theta} = \theta$, for every influence matrix satisfying Assumption 1 and every profile $(\lambda_1, \dots, \lambda_m)$.*

Proof. Substituting $\vec{c} = \theta^{-1}\mathbf{1}$ in row a of (5) and using $\sum_j A_{aj} = 1$ gives $(\theta + \lambda_a)/\theta - \lambda_a/\theta = 1$. Uniqueness follows from strict diagonal dominance, as above. $\square$

The hypothesis is that the product $\alpha_a(1 - \gamma_a)$ be common, so neutrality holds on the whole level set $\{(\alpha, \gamma) : \alpha(1 - \gamma) = \theta\}$. Along that set an investor with a stronger comparison motive is also more risk averse in own wealth, in exactly the proportion that cancels. That is the content of the theorem, and it is weaker than it sounds if θ_a is read as the conventional coefficient of absolute risk aversion, which by Definition 1 it is not.

The cancellation occurs row by row, before any aggregation, which is why nothing about A or about the dispersion of λ_a enters. Economically, the benchmark is fixed in a unilateral deviation: under Assumption 2 the positions of others do not respond. Neutrality arises instead because at the equilibrium profile the benchmark carries the same risky exposure as the investor himself, so that $(\theta + \lambda_a)\vec{\pi} - \lambda_a A \vec{\pi} = (\theta + \lambda_a)\vec{\pi} - \lambda_a \vec{\pi} = \theta \vec{\pi}$. The benchmark exposure offsets the additional λ_a loading on own equilibrium wealth in the level of total risky exposure, not through any conjectured response to the deviation.

Theorem 1 is the network extension of the CARA case of [12], and we claim no more for it than that. His Proposition 2 identifies common cautiousness — here, common θ — as what makes the weighting drop out; the addition is that the weighting may be agent-specific and the comparison intensities may differ, and neither disturbs the conclusion.

Remark 2 (Comparison at constant own-wealth risk aversion). Theorem 1 varies λ_a holding θ fixed. The other natural path holds α fixed and varies γ_a , so that $\theta_a = \alpha(1 - \gamma_a)$ is dispersed whenever γ_a is. Neutrality then fails, but not through a new mechanism: the demand scales solve (5) with $\theta_a = \alpha(1 - \gamma_a)$ and $\lambda_a = \alpha\gamma_a$, and dispersion of the keeping-up weight is dispersion of net risk aversion. The two paths are therefore two coordinates of one statement. What neutrality requires is that $\alpha_a(1 - \gamma_a)$ be common; what breaks it is dispersion in that product, however generated.

Gómez's Proposition 2 requires more, in a second way. It asks that the preference parameters be identical across agents and, barring uniform weights, that the equilibrium value of the endowment be identical as well. Initial wealth is absent from (4) altogether, since CARA demands have no wealth effects, so no restriction on the endowment distribution is needed here.

Remark 3 (Two notions of neutrality). Theorem 1 asserts that individual positions are unchanged, which is stronger than saying the effective coefficient is unchanged. The latter requires only that $\sum_a c_a$ be unchanged, and deviations can offset. The distinction is not hypothetical: it occurs inside the example of Proposition 2, whose $\bar{\theta}(\lambda)$ returns to its no-comparison value at $\lambda = \frac{1}{3}$. There

$$\vec{c}(0) = (2, 1, \tfrac{1}{2}), \quad \vec{c}(\tfrac{1}{3}) = (\tfrac{5}{3}, \tfrac{7}{6}, \tfrac{2}{3}), \quad \bar{\theta}(0) = \bar{\theta}(\tfrac{1}{3}) = \tfrac{6}{7},$$

so the aggregate is exactly neutral while individual positions move by -16.7 , $+16.7$ and $+33.3$ per cent. Results below concern $\bar{\theta}$, so the weaker notion is the relevant one, and statements about it should not be read as statements about allocations.

5 Heterogeneous net risk aversion

We now let θ_a vary, which is exactly the configuration in which the aggregation conditions of [12] fail. The system (5) still has a unique positive solution, but it is no longer proportional to $\mathbf{1}$, and $\bar{\theta}$ depends on λ and on A .

Throughout this section $\lambda_a = \lambda$ is common, so that λ indexes a one-parameter family of economies running from no comparison to complete comparison. Two limits of that family can be computed in closed form for *every* network, and between them they organise everything that follows. Write

$$\vec{t} = \mathbb{D}_\theta^{-1} \mathbf{1}, \quad t_a = 1/\theta_a, \tag{7}$$

for the vector of risk tolerances, so that $\vec{c}(0) = \vec{t}$ and $\bar{\theta}(0) = H(\theta)$ is the harmonic mean of net risk aversion.

5.1 The marginal effect of comparison

Theorem 2. *Let Assumption 1 hold and $\lambda_a = \lambda$. Then*

$$\bar{\theta}'(0) = \frac{m \bar{t}^\top (\mathbb{I} - A) \bar{t}}{(\mathbf{1}^\top \bar{t})^2}. \quad (8)$$

If in addition A is doubly stochastic — not necessarily symmetric — then

$$\bar{t}^\top (\mathbb{I} - A) \bar{t} = \frac{1}{2} \sum_{a,j} A_{aj} (t_a - t_j)^2 \geq 0, \quad (9)$$

so $\bar{\theta}'(0) \geq 0$, with equality if and only if $t_a = t_j$ whenever $A_{aj} > 0$. For A merely row stochastic the sign of (8) is unrestricted.

Proof. Write $M(\lambda) = \mathbb{D}_\theta + \lambda(\mathbb{I} - A)$, so that $\bar{c} = M^{-1} \mathbf{1}$ and $\bar{c}' = -M^{-1}(\mathbb{I} - A)\bar{c}$. At $\lambda = 0$ we have $M(0) = \mathbb{D}_\theta$ and $\bar{c}(0) = \bar{t}$, hence $\mathbf{1}^\top \bar{c}'(0) = -\mathbf{1}^\top \mathbb{D}_\theta^{-1}(\mathbb{I} - A)\bar{t} = -\bar{t}^\top (\mathbb{I} - A)\bar{t}$, using $\mathbf{1}^\top \mathbb{D}_\theta^{-1} = \bar{t}^\top$. Since $\bar{\theta} = m/(\mathbf{1}^\top \bar{c})$, $\bar{\theta}' = -m(\mathbf{1}^\top \bar{c}')/(\mathbf{1}^\top \bar{c})^2$, which gives (8). For the second claim, row and column sums of A both equal one, so $\sum_{a,j} A_{aj}(t_a^2 + t_j^2)/2 = \frac{1}{2}(\sum_a t_a^2 + \sum_j t_j^2) = \bar{t}^\top \bar{t}$, and therefore $\frac{1}{2} \sum_{a,j} A_{aj}(t_a - t_j)^2 = \bar{t}^\top \bar{t} - \bar{t}^\top A \bar{t}$, which is (9). The last claim follows from the example of Proposition 2, whose column sums are $(2, 1, 0)$. $\square$

Equation (9) is a Dirichlet form: the marginal effect of comparison is a weighted sum of squared differences in risk tolerance across the links of the network. It is positive exactly when investors look at somebody unlike themselves, and the relevant notion of “unlike” is difference in tolerance, not in aversion. Double stochasticity is what makes the quadratic form a Dirichlet form; economically it says that attention is balanced, each investor receiving as much of it as he gives. When attention is unbalanced the identity fails and the sign is free, which is the content of the counterexample below.

5.2 The limit of intense comparison

The other end of the family is governed by the stationary distribution of A read as a Markov chain. Let $C_1, \dots, C_r$ be its recurrent classes, $\bar{\pi}^{(k)}$ the stationary distribution on C_k normalised to sum to one, and $h_k(a)$ the probability that a chain started at a is absorbed into C_k .

Lemma 1. $c_a \leq 1/\min_b \theta_b$ for every a and every $\lambda \geq 0$.

Proof. Let $c^* = c_{a^*}$ be the largest entry. Row a^* of (5) reads $(\theta_{a^*} + \lambda)c^* = 1 + \lambda \sum_j A_{a^*j} c_j \leq 1 + \lambda c^*$, so $\theta_{a^*} c^* \leq 1$. $\square$

Theorem 3. *Let Assumption 1 hold and $\lambda_a = \lambda$. Then*

$$\lim_{\lambda \rightarrow \infty} \bar{\theta}(\lambda) = \frac{m}{\sum_{k=1}^r \frac{\mathbf{1}^\top \bar{h}_k}{\bar{\pi}^{(k)\top} \bar{\theta}}}. \quad (10)$$

In particular, if A has a single recurrent class with stationary distribution $\vec{\pi}$, then $\bar{\theta}(\lambda) \rightarrow \vec{\pi}^\top \vec{\theta}$.

Proof. The kernel of $\mathbb{I} - A$ is spanned by the absorption vectors $\vec{h}_1, \dots, \vec{h}_r$, which satisfy $\sum_k \vec{h}_k = \mathbf{1}$. Dividing $M(\lambda)\vec{c} = \mathbf{1}$ by λ gives $(\lambda^{-1}\mathbb{D}_\theta + \mathbb{I} - A)\vec{c} = \lambda^{-1}\mathbf{1}$, and $\vec{c}$ is bounded by Lemma 1, so any limit point $\vec{c}_\infty$ satisfies $(\mathbb{I} - A)\vec{c}_\infty = 0$, that is $\vec{c}_\infty = \sum_k \kappa_k \vec{h}_k$. To identify κ_k , extend $\vec{\pi}^{(k)}$ by zero off C_k and left-multiply $M(\lambda)\vec{c} = \mathbf{1}$ by $\vec{\pi}^{(k)\top}$. Since C_k is closed, $\vec{\pi}^{(k)\top} A = \vec{\pi}^{(k)\top}$, so the term in λ vanishes identically and $\vec{\pi}^{(k)\top} \mathbb{D}_\theta \vec{c} = 1$ for every λ . Passing to the limit and using $h_l = \delta_{kl}$ on C_k gives $\kappa_k \vec{\pi}^{(k)\top} \vec{\theta} = 1$. Then $\mathbf{1}^\top \vec{c}_\infty = \sum_k \kappa_k \mathbf{1}^\top \vec{h}_k$, and $\bar{\theta} = m/(\mathbf{1}^\top \vec{c})$ gives (10). The limit point is unique, so the whole family converges. $\square$

Theorem 3 is the organising result of the paper. Under intense comparison the aggregate is no longer the harmonic mean of net risk aversion but a *stationary* average: influence, not population, does the weighting. Reading $\bar{\theta}(0) = H(\theta)$ against (10) gives at once a sufficient condition for the comparison to lower the effective coefficient,

$$\vec{\pi}^\top \vec{\theta} < H(\theta) \implies \bar{\theta}(\lambda) < \bar{\theta}(0) \text{ for all large } \lambda, \quad (11)$$

in the single-class case. Reversal is therefore not an artefact of small examples but a statement about the correlation between net risk aversion and stationary influence: it occurs when the investors who attract attention in the long run are less risk averse than the population as a whole. The three results that follow are the three ways (10) can turn out.

5.3 The mean-field limit

Proposition 1. *Let $\lambda_a = \lambda$ for all a and consider a sequence of economies indexed by m in which the empirical distribution of θ_a converges weakly to a law with $\mathbb{E}[(\theta + \lambda)^{-1}] < \infty$ and $\{(\theta + \lambda)^{-1}\}$ uniformly integrable. Suppose the networks satisfy*

$$\theta_a \geq \underline{\theta} > 0 \text{ for all } a, m, \quad \delta_m = \max_a \left\| A_a - \frac{1}{m} \mathbf{1} \right\|_1 \longrightarrow 0. \quad (12)$$

Then, writing $\bar{\theta}_m$ for the effective coefficient of the m -investor economy,

$$\bar{\theta}_m(\lambda) \longrightarrow \bar{\theta}_\infty(\lambda) = H(\theta + \lambda) - \lambda, \quad H(x) = (\mathbb{E}[1/x])^{-1}. \quad (13)$$

The limit $\bar{\theta}_\infty$ is differentiable wherever the moments below exist, and

$$\frac{d\bar{\theta}_\infty}{d\lambda} = \frac{\text{Var}[(\theta + \lambda)^{-1}]}{\mathbb{E}[(\theta + \lambda)^{-1}]^2} \geq 0, \quad (14)$$

with equality if and only if θ is degenerate. If moreover $\mathbb{E}[\theta^3] < \infty$ then $\bar{\theta}_\infty = \mathbb{E}[\theta] - \text{Var}(\theta)/\lambda + O(\lambda^{-2})$ as $\lambda \rightarrow \infty$. Equation (14) is the derivative of the limit; we do not claim it is the limit of the derivatives of the finite economies.

Proof. By Lemma 1, $\|\vec{c}\|_\infty \leq 1/\underline{\theta}$ uniformly in m and λ . Hence $|\sum_j A_{aj}c_j - \bar{c}_m| = |(A_{a\cdot} - m^{-1}\mathbf{1})^\top \vec{c}| \leq \delta_m \|\vec{c}\|_\infty \leq \delta_m/\underline{\theta} \rightarrow 0$ uniformly in a , which is the mixing condition in endogenous form and is where the uniform bound is needed. Under it, row a of (5) becomes $(\theta_a + \lambda)c_a = 1 + \lambda\bar{c}$ in the limit, so $c_a = (1 + \lambda\bar{c})/(\theta_a + \lambda)$. Averaging and writing $M = \mathbb{E}[(\theta + \lambda)^{-1}]$ gives $\bar{c} = M/(1 - \lambda M)$ and $\bar{\theta} = 1/M - \lambda$. Differentiating gives (14), whose sign is Cauchy–Schwarz, with equality only for degenerate θ . The expansion follows from $\mathbb{E}[(\theta + \lambda)^{-1}] = \lambda^{-1}(1 - \mathbb{E}[\theta]/\lambda + \mathbb{E}[\theta^2]/\lambda^2 + O(\lambda^{-3}))$, which requires the third moment to control the remainder. $\square$

In this limit comparison interpolates the aggregation of risk aversion between the two classical aggregators: the harmonic mean at $\lambda = 0$, which is the standard CARA result, and the arithmetic mean as $\lambda \rightarrow \infty$. Since $H(\theta) \leq \mathbb{E}[\theta]$, the largest increase in the effective coefficient available is the harmonic–arithmetic gap of the net risk-aversion distribution, a statistic of the cross-section rather than a free parameter.

Condition (12) is primitive: it constrains only A and the support of θ . The leave-one-out network $A_{aj} = 1/(m-1)$, $j \neq a$, has $\delta_m = 2/m$ exactly, so it qualifies; exact equality at finite m would require identical rows, which Assumption 1 forbids since $A_{aa} = 0$. More generally δ_m is twice the total-variation distance of each row from the uniform distribution, so (12) asks that every investor’s benchmark be close to the population average — which is what “mean field” should mean, and is checkable from the network alone. The uniform lower bound $\underline{\theta}$ cannot be dropped: it is what converts closeness of rows into closeness of benchmarks, since $\vec{c}$ is controlled only through Lemma 1.

Remark 4 (Consistency with Theorem 3). Condition (12) forces the stationary distribution towards the uniform one, and (10) then returns $\vec{\pi}^\top \vec{\theta} \rightarrow \mathbb{E}[\theta]$, which is the $\lambda \rightarrow \infty$ end of (13). The mean-field regime is thus the case of Theorem 3 in which influence is spread evenly, and $\bar{\theta}'(0) \geq 0$ in it because near-uniform rows are near doubly stochastic, so (9) applies approximately. Both directions of the interpolation are consequences of the two limits.

5.4 Non-monotonicity and sign reversal

Proposition 2. *There exist normalised benchmarks and net risk-aversion profiles for which $d\bar{\theta}/d\lambda < 0$ on a non-empty interval and for which $\bar{\theta}(\lambda) < \bar{\theta}(0)$ at sufficiently large comparison intensities.*

Proof. Take $m = 3$, $\theta = (\frac{1}{2}, 1, 2)$, $\lambda_a = \lambda$ and

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

which satisfies Assumption 1. Solving (5) gives

$$\bar{\theta}(\lambda) = \frac{3(\lambda + 2)(3\lambda + 1)}{12\lambda^2 + 24\lambda + 7}, \quad \bar{\theta}'(\lambda) = -\frac{3(12\lambda^2 + 6\lambda - 1)}{(12\lambda^2 + 24\lambda + 7)^2}. \quad (15)$$

Hence $\bar{\theta}'(0) = 3/49 > 0$ and $\bar{\theta}'(\lambda) < 0$ for $\lambda > (\sqrt{21} - 3)/12 \approx 0.132$. The function rises from $\bar{\theta}(0) = 6/7 \approx 0.857$ to a maximum ≈ 0.860 at that point, then falls, returning to $6/7$ at $\lambda = 1/3$ and continuing downwards: $\bar{\theta}(1) = 0.837209$ and $\bar{\theta}(2) = 0.815534$. $\square$

Both features of the example are read off the two limits of Section 5.1 and Section 5.2, and neither depends on the closed form. The column sums of A are $(2, 1, 0)$, so A is not doubly stochastic and (9) does not apply; (8) nonetheless returns $\bar{\theta}'(0) = 3/49 > 0$, so comparison initially raises the effective coefficient. At the other end, investor 3 is transient and $\{1, 2\}$ is the unique recurrent class, with $\vec{\pi} = (\frac{1}{2}, \frac{1}{2}, 0)$. Theorem 3 therefore gives

$$\lim_{\lambda \rightarrow \infty} \bar{\theta}(\lambda) = \vec{\pi}^\top \vec{\theta} = \frac{1}{2}(\frac{1}{2} + 1) = \frac{3}{4} < \frac{6}{7} = H(\theta) = \bar{\theta}(0),$$

so (11) holds and reversal is forced. The economics is now explicit. The most risk averse investor watches the least risk averse and is watched by nobody, so stationary influence assigns him weight zero: his risk aversion is present in the harmonic mean at $\lambda = 0$ and absent from the stationary mean at $\lambda = \infty$. Non-monotonicity is the collision of the two limits — a positive marginal effect at the origin with a strictly lower value at infinity — and the closed form only locates the turning point.

How often reversal occurs is a question about a distribution over networks, and the answer is a property of that distribution rather than of the mechanism. Under the dense i.i.d.-weight design specified below it becomes rare as the number of investors grows: the rows of A approach the uniform distribution, so by Remark 4 the design converges to the mean-field case, in which (9) applies and the effect is non-negative. Nothing in this licenses a claim about networks in general. Real influence structures carry communities, hubs and correlation between characteristics and position, and Remark 5 shows that reversal survives at any population size once such structure is present. Table 1 reports two distinct frequencies over 20,000 independent draws at each size: the *cumulative* event $\bar{\theta}(0.5) < \bar{\theta}(0)$, and the *marginal* event $\bar{\theta}'(0.5) < 0$. The derivative is computed analytically rather than by finite differences: writing $M(\lambda) = \mathbb{D}_\theta + \lambda(\mathbb{I} - A)$ and $\vec{c} = M^{-1}\mathbf{1}$,

$$\vec{c}' = -M^{-1}(\mathbb{I} - A)\vec{c}, \quad \bar{\theta}' = -\frac{m \mathbf{1}^\top \vec{c}'}{(\mathbf{1}^\top \vec{c})^2}. \quad (16)$$

The number of investors is held fixed within each row, so that the comparison across rows identifies the effect of size. Off-diagonal entries of A are uniform on $(0, 1)$ before row normalisation, and $\theta_a = \max\{1 + \sigma Z_a, 0.15\}$ with Z_a standard normal.

Remark 5 (Reversal at any population size). The decay in Table 1 reflects mixing, not size. Because (5) is local, a block-diagonal A decouples: each block solves its own system and $\bar{\theta} = (m^{-1} \sum_a c_a)^{-1}$ is the reciprocal of the population-average capacity. Replicating an identical block leaves that average, and hence $\bar{\theta}$, unchanged. Taking k disjoint copies of the network of Proposition 2 therefore gives $\bar{\theta}(\lambda)$ independent of k , so that $\bar{\theta}(1) = 0.837209 < 6/7 = \bar{\theta}(0)$ at $m = 3k$ for every k ; we verify this numerically up to $m = 3000$. The construction is robust to weak ties across blocks: mixing each row with a fraction ϵ of the uniform complete network preserves the reversal at $m = 30$ for

m	σ	cumulative: $\bar{\theta}(0.5) < \bar{\theta}(0)$		marginal: $\bar{\theta}'(0.5) < 0$	
		frequency	95% interval	frequency	95% interval
3	0.4	0.0919	[0.0879, 0.0959]	0.0952	[0.0912, 0.0993]
3	0.8	0.0340	[0.0316, 0.0366]	0.0372	[0.0346, 0.0399]
6	0.4	0.0057	[0.0047, 0.0068]	0.0062	[0.0052, 0.0073]
6	0.8	0.0008	[0.0005, 0.0014]	0.0008	[0.0005, 0.0012]
10	0.4	0.0001	[0.0000, 0.0004]	0.0001	[0.0000, 0.0004]
10	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
20	0.4	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
20	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
60	0.4	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
60	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]

Table 1: Frequency of negative cumulative and marginal effects at $\lambda = 0.5$, over 20,000 draws per row with Wilson intervals. All ten cells of the design are reported; entries of 0.0000 are exact zeros, no draw out of 20,000 producing the event, and 0.0002 is the upper endpoint of the 95% Wilson interval when no event is observed. Under this design reversal is already below one per cent at $m = 6$ and indistinguishable from zero at $m \geq 10$. The decay in m is specific to this ensemble; see Remark 5. Reversal is also rarer at $\sigma = 0.8$, but the two rows are not a clean comparative static in dispersion: the truncation at 0.15 binds for 14.4 per cent of draws at $\sigma = 0.8$ against 1.7 at $\sigma = 0.4$, which moves the shape of the distribution and raises $\mathbb{E}[\theta]$ from 1.00 to 1.06. We report both rows and draw no conclusion from their difference.

$\epsilon \leq 0.1$ ($\bar{\theta}(1) = 0.8526$ at $\epsilon = 0.1$) and destroys it by $\epsilon = 0.2$ ($\bar{\theta}(1) = 0.8681$). Table 1 should therefore be read as describing a dense ensemble in which local benchmarks approach a common average, and not as a claim about large networks in general: block structure, or any persistent correlation between type and network position, sustains the reversal at arbitrary population size.

Note the contrast with Proposition 3, which uses the same block structure to the opposite end. There net risk aversion is constant *within* blocks and the mechanism vanishes; here it is heterogeneous within blocks and absent *across* them, and the mechanism survives at any scale. What separates the two is where the heterogeneity sits relative to the partition, not the topology.

5.5 Perfect homophily blocks the mechanism

Proposition 3. *Partition the investors into groups such that $A_{aj} = 0$ whenever a and j belong to different groups, and suppose θ_a is constant within each group. Then $c_a = 1/\theta_a$ solves (5) for every profile $(\lambda_1, \dots, \lambda_m)$, and $\bar{\theta}$ equals the harmonic mean of θ regardless of the intensity of comparison.*

Proof. Row a of (5) reads $(\theta_a + \lambda_a)c_a - \lambda_a \sum_j A_{aj}c_j = 1$. If $c_j = 1/\theta_j$ and $\theta_j = \theta_a$ for every j with $A_{aj} > 0$, the left-hand side is $(\theta_a + \lambda_a)/\theta_a - \lambda_a/\theta_a = 1$. Uniqueness gives the claim, and $\bar{\theta} = m/\sum_a(1/\theta_a)$ is the harmonic mean of θ . $\square$

Theorem 3 explains why. Each group is a recurrent class, so $r > 1$, and θ is constant on C_k , whence $\vec{\pi}^{(k)\top} \vec{\theta} = \theta_k$ whatever the stationary distribution inside the class. With every investor in some class, $\mathbf{1}^\top \vec{h}_k$ is the size m_k of the class and (10) reads $m/\sum_k m_k/\theta_k = H(\theta)$. The two limits therefore coincide, which is the strongest possible form of neutrality of the aggregate: not that $\bar{\theta}$ returns to $H(\theta)$, but that it never leaves. With two equally sized groups at $\theta \in \{0.5, 2\}$, $\bar{\theta} = 0.8$ at $\lambda = 0$, at $\lambda = 1$ and at $\lambda = 1000$; the arithmetic mean 1.25 that Proposition 1 would predict is never approached.

The hypothesis is exact segregation together with exact constancy of θ within blocks, and both are needed: if θ varies inside a class then $\vec{\pi}^{(k)\top} \vec{\theta} \neq \theta_k$ and (10) moves, as Remark 5 exploits. So the correct statement is that *perfect homophily by net risk-aversion type* suppresses the mechanism, not that homophily does. It is worth knowing because assortativity on wealth and on financial sophistication is a documented feature of social networks [15, 6].

6 Conclusion

Relative wealth concerns in a CARA economy aggregate through the influence network in a way that two limits describe completely. The marginal effect at zero comparison is a quadratic form in risk tolerances, which is a Dirichlet form — and so non-negative — exactly when attention is balanced. The limit under intense comparison is a stationary average of individual coefficients, weighted by influence rather than by population.

Between them these two statements account for the mean-field interpolation from the harmonic to the arithmetic mean, for non-monotonicity and sign reversal, and for the complete inertness of the aggregate under perfect homophily by type. What decides which case obtains is the correlation between net risk aversion and stationary influence.

Two boundaries of the result should be kept in view. It concerns the effective market coefficient, not asset prices: the model delivers a mean–covariance relation conditional on the covariance matrix, and closing it into a full one-period economy with exogenous quantities and endogenous prices is the obvious next step. And θ_a is risk aversion net of the comparison, not the coefficient of absolute risk aversion in own wealth, so “common risk aversion” in Theorem 1 means a common value of the product $\alpha_a(1 - \gamma_a)$, along which stronger comparison motives come paired with greater own-wealth risk aversion.

The practical reading is that a representative-agent treatment of relative concerns reproduces the mean-field case and is safe only where influence is close to evenly spread. Where attention is unbalanced — where some investors are watched much more than they watch — it can err in sign. Two questions follow naturally. The first is quantitative: how far can $\vec{\pi}^\top \vec{\theta}$ fall below $H(\theta)$ for networks with realistic degree distributions and realistic correlation between risk aversion and centrality? The second is empirical: stationary influence is in principle measurable, and Theorem 3 converts it into a prediction about the aggregate.

Replication

All numerical results are reproducible from the repository

<https://github.com/renatovicente/price-of-risk-aggregation>

The module `scripts/network.py` solves (5); `scripts/neutrality.py` verifies Theorem 1 and Proposition 1; `scripts/sign_reversal.py` produces Table 1, including the network and type generators and the seeds; `scripts/block_reversal.py` produces the block replication and the mixing robustness of Remark 5; and `scripts/characterization.py` checks Theorems 2 and 3, the Dirichlet identity (9), the bound of Lemma 1 and each of the three cases of (10). The scripts depend only on `numpy` and `scipy`, read no data and draw from fixed seeds; `./reproduce.sh` runs all four in about fifteen seconds.

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